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Acids, Bases and Salts

There are different chemical Elements known to us at present, these Elements combined to form a large number of compounds, on the basis of their chemical properties these are :-

1. Acid
2. Bases
3. Salts

Q How we came to know that wheather a substance is an Acid or Base ?

Ans:- Well, We use some chemical substances which tells us wheather a substance we have taken is an acid or Base, By changing its physical or chemical properties like (smell), colour etc.

★ Indicators

An indicator is a chemical substance which tell's us wheather the chemical we are taking is an acid or Base by changing its colour.

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1. Natural Indicators :

Indicators which are obtained naturally are called natural indicators

eg:- Litmus paper, Red cabbage, Turmeric, onion etc.

Litmus

An Acid turn Blue Litmus to Red
A Base turn Red Litmus to Blue

Red Cabbage

Red cabbage remains Red in Acid
Turns green in Basic solution

Turmeric

Remains yellow in Acid
Becomes Red-Brown in Base

2. Olfactory Indicators :-

Substances which change their odour (smell) with Acid and Bases are called olfactory indicators

eg:- Vanilla, Onion, clove etc.

Onion

Onion lost its smell in Basic solution.
Restore its smell in Acidic solution.

Vanilla

Loose its smell in Basic solution.
Restore its smell in Acidic solution.

Synthetic Indicators:

Synthetic indicators are obtained from Artificial substances that are produced in Lab's
eg:- Methyl orange, Phenolphthalein.

Methyl orange
Turns Red in Acidic solution.
yellow in Basic solution.

Phenolphthalein
Colorless in Acidic solution.
Pink colour in Basic solution.

Acids :-

Physical Properties

★ Acids have a sour taste.
★ They can change colour of Blue Litmus to Red.
★ Acids gets Ionize on dissolving in water and Produce (H^+) Ion also known as Hydrogen Ion.

Organic Acids

★ Acids present in plants and animals are organic / weak Acids.
Ex: Acetic Acid, Lactic Acid, Tartaric Acid
★ The Weak Acid is not harmful to Eat or Drink.
★ They cannot Ionize water completely.

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Minerals Acid :

- Acids prepared From the minerals of Earth are called Mineral Acids.
  Eg: Hydrochloric Acid, Sulphuric Acid and Nitric Acid.
- Concentrated mineral Acids are very dangerous they can burn our hands and clothes.
- These Acids should be handle with care and generally mixed with water to dilute them such Acids are called diluted Acids.
- During dilution of an Acid, Acid should be always mixed with water, Not water in Acid that can cause splash on your Face and can cause severe Acid burns.
- They can Ionize water completely.

Sulphuric acid $\longrightarrow H_2SO_4$ $[2H^+ \quad SO_4^{2-}]$

Nitric acid $\longrightarrow HNO_3$ $[H^+ \quad NO_3^-]$

Hydrochloric acid $\longrightarrow HCl$ $[H^+ \quad Cl^-]$

Carbonic acid $\longrightarrow H_2CO_3$ $[2H^+ \quad CO_3^{2-}]$

Acetic acid $\longrightarrow CH_3COOH$ $[H^+ \quad CH_3COO^-]$

The positive ion of all acids are same (H^+) which is responsible for Acidic properties

Chemical Properties of Acids

• Reaction of acid with metal

Acid react with metal to form salt corresponding to metal and hydrogen gas.

Metal + Acid ⟶ Salt of Metal + Hydrogen gas ↗ Pop sound

eg: $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2\uparrow$

$Ca + 2HCl \longrightarrow CaCl_2 + H_2\uparrow$

$Fe + HCl \longrightarrow FeCl_2 + H_2\uparrow$

★ When a burning candle is brought near hydrogen gas, it will burn with pop sound and candle will blows off.

★ As we can see that acids reacts with metal to form poisonous salt, therefore we do not keep curd or sour eatables (acidic substances) in brass or copper (metals).

• Reaction of Acid with Base

Acids reacts with Base to form salt and water.

• This is called Neutralisation Reaction

Acid + Base ⟶ Salt + Water

eg: $HCl + NaOH \longrightarrow NaCl + H_2O$

★ Reaction of Acid With Metal oxide

Acids reacts with Metaloxide to form salt and water similar to an Acid Base Reaction.

Metal oxide + Acid ⟶ Salt + Water

eg: $CuO + 2HCl \longrightarrow CuCl_2 + H_2O$

$Na_2O + 2HCl \longrightarrow 2NaCl + H_2O$

★ Metal oxides are basic in nature.

★ Reaction of Acid with Metal Carbonate or metal hydrogen Carbonate.

- Acids react with Metal carbonate / Metal Hydrogen carbonate to form salt + Water + carbondioxide gas.

Metal carbonate / Metal hydrogen carbonate + Acid ⟶ Salt + CO_2 + H_2O

eg: $CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2\uparrow$
calcium carbonate

$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2\uparrow$
Sodium carbonate

$NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2\uparrow$
Sodium hydrogen carbonate

Bases

Physical properties

★ Bases are those chemical substances which have bitter taste.

★ Bases produce Hydroxide ions (OH^-) when dissolve in water.

★ A Base can neutralise an Acid.

★ Bases Feel soapy to touch.

★ A base which is soluble in water is called an Alkali.

★ Base which completely ionize in water and thus produce Large amount of hydroxide ions is called a strong base or strong Alkali
eg:- Sodium Hydroxide.

★ Base which partially ionize in water and thus produce small amount of hydroxide ions is called a weak base or weak Alkali.
eg:- Magnesium hydroxide.

★ Potassium Hydroxide (KOH) $[K^+ \quad OH^-]$
Calcium Hydroxide $Ca(OH)_2$ $[Ca^{+2} \quad 2OH^-]$
Magnesium Hydroxide $Mg(OH)_2$ $[Mg^{2+} \quad 2OH^-]$
Sodium Hydroxide $NaOH$ $[Na^+ \quad OH^-]$
Aluminium Hydroxide $Al(OH)_3$ $[Al^{3+} \quad 2OH^-]$

} Here OH^- in all bases are same

Chemical Properties of Bases

★ Reaction of Base with metal

Bases react with metal to form salt corresponding to metal and hydrogen gas.

Base + Metal ⟶ Salt of metal + Hydrogen(g)

eg:- $Zn + 2NaOH \longrightarrow Na_2ZnO_2 + H_2\uparrow$

Note Bases को भी metal container में नहीं रखते।

[redacted]

★ Reaction of Base with Non-Metal oxide

Bases react with non-metallic oxide to form salt and water.

Base + non-metallic oxide ⟶ Salt + Water
(Acidic)

eg:- $CO_2 + Ca(OH)_2 \longrightarrow CaCO_3 + H_2O$

⇒ This reaction proves that non-metallic oxides are acidic in nature.

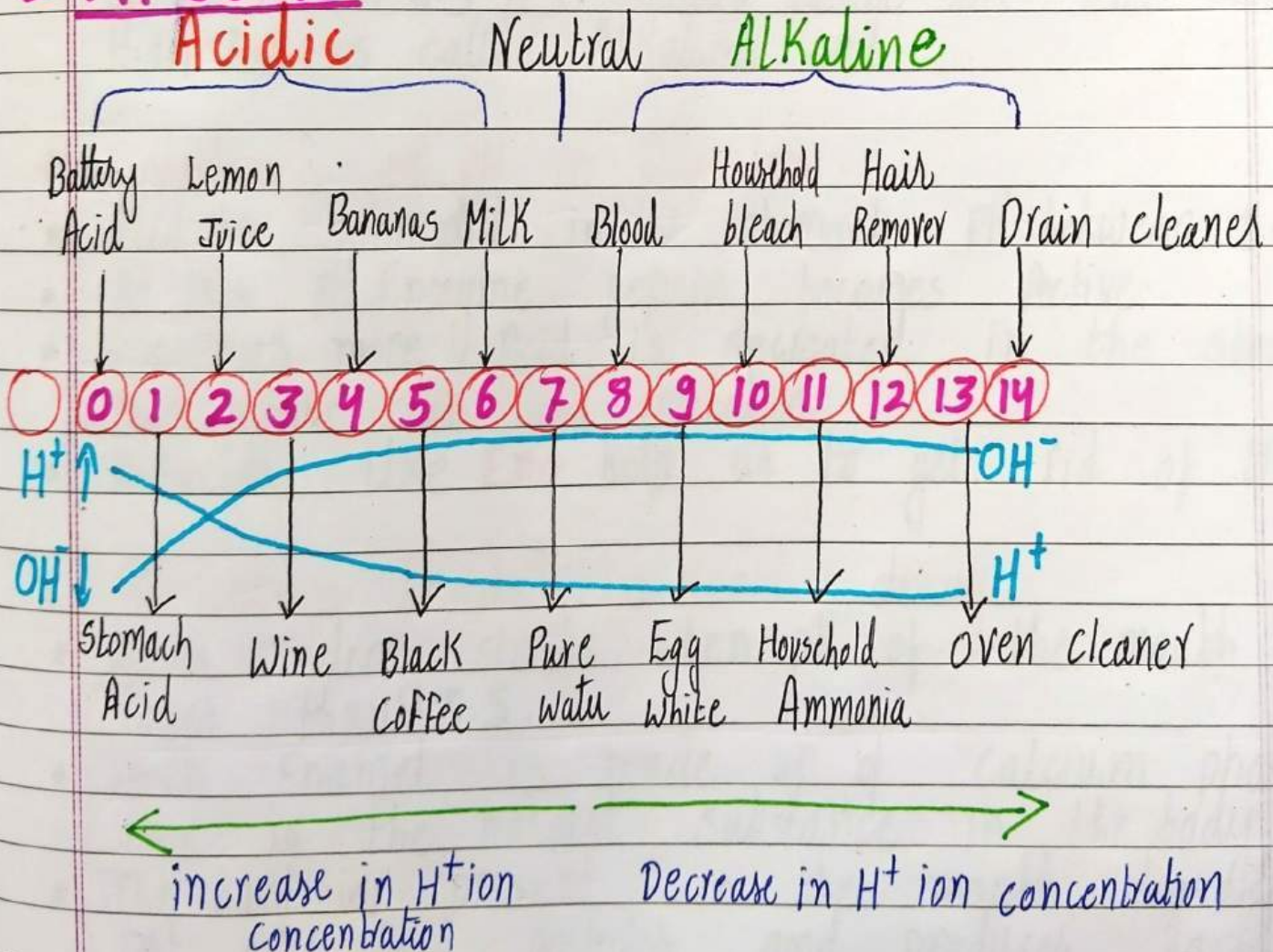
⇒ What happens to an Acid or a Base in a water solution.

When an Acid or Base is dissolve in water, they get dissociated into ions.

$$HCl + H_2O \longrightarrow H^+ + Cl^-$$

$$NaOH + H_2O \longrightarrow Na^+ + OH^-$$

= PH Scale



★ Importance of PH in everyday Life.

★ Importance of pH in existing of a Living being

- our body works within the pH range of 7.0 - 7.8
- When pH of rain water is Less than 5.6 it is called Acid rain.

★ Importance of pH in Soil.

- Soil whose pH is between 6.5 - 7.3 then its growth and development is good.
- If Less than 6.5 its called acidic soil and more than 7.3 is called Alkaline soil.

★ Importance of pH in Digestion.

- HCl is secreated in the stomach, pH between 1 and 3.
- At this pH, Enzyme pepsin becomes Active.
- Sometimes more Acid is secreated in the stomach this cause pain and irritation.
- Antacids Like **Eno** help us to get rid of this.

★ PH change Leads to tooth decay

- Tooth decay starts when pH of the mouth is lower than 5.5.
- Tooth Enamel is made up of calcium phosphate and is the hardest substance in the body.
- The bacteria present in the mouth degrade the Left over food particles and produces acid.
- To protect tooth decay, toothpaste which are basic in nature are used to neutralize the Excess acid.

★ Self defence by Animals & plants through chemical welfare:

- When insects like honeybee, ant etc, bite they inject an acid into the skin that causes pain and irritation. If a mild base like baking soda is applied to affected area.

★ pH in plants:

- Stinging hair of nettle leaves injects methanoic acid in the skin which causes burning pain.
- It is cured by rubbing the affected area with the leaves of dock plant, which often grows besides the nettle plant.

Dilution

Mixing an acid or base with water result in decrease in the concentration of ions (H_3O^+/OH^-) per unit volume. Such a process is called dilution and the acid or the base is said to be diluted.

⇒ Imp

The acid must always be added slowly to water with constant stirring because the process is highly exothermic. If water is added to a concentrated acid, the heat generated may cause the mixture to splash out and cause burns. The glass container may also break due to excessive heating.

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★★ Some naturally occuring acids.

Natural source	Acid
• Vinegar	Acetic acid
• Orange	Citric acid
• Tamarind	Tartaric acid
• Tomato	oxalic acid
• Curd	Lactic acid
• Lemon	cibric acid
• Ant sting	Methanoic acid
• Nettle Sting	Methanoic acid.

SALTS

★ A salt is a compound Formed when Acid react with bases. (Neutralisation reaction).

★ The nature of the salt can be:

1. Acidic :- The salt is made by the reaction of **strong Acid** and **weak base**.
   Eg:- NH_4Cl
2. Basic :- The salt is made by the reaction of **strong Base** and **weak Acid**.
   Eg:- Na_2CO_3
3. Neutral :- The salt is made by the reaction of **strong Base** and **strong Acid**.
   Eg:- NaCl

Common Salt [Sodium Chloride, NaCl]:-

This is the salt we use in food. Seawater contains many salts dissolved in it. Sodium chloride is separated from these salts.

Formation (in Laboratory)

Sodium chloride is formed by the reaction of sodium hydroxide and HCl.

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

Uses

* Used in cooking food.
* Used as a preservative
* Used to melt ice which collects on Roads in cold areas.
* Used in manufacturing of soap.
* Used in chemical Industry.

* Sodium Hydroxide

* When Electricity is passed through an Aqueous solution of sodium chloride (Brine) it decomposes to form Sodium Hydroxide.

This is also called "Chlor-Alkali Process".

$$2NaCl(aq) + 2H_2O(l) \longrightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$$

Here,
Chlorine gas is given off at the anode and hydrogen gas at cathode. and sodium hydroxide solution is formed near cathode.

Different uses of these products.

H_2 ⟶ Fuels, making of Ammonia for fertilizers

Cl_2 ⟶ Water treatment, Pesticides, cleaning of swimming pool.

NaOH ⟶ Soaps, detergents, De-greasing metals, paper making.

Bleaching Powder [Calcium oxychloride, $CaOCl_2$]

- Bleaching powder is produced by the Action of chlorine on dry slacked lime.

$$Ca(OH)_2 + Cl \longrightarrow CaOCl_2 + H_2O$$

Uses:

★ For bleaching cotton and linen in textile industries.
★ As an oxidising Agent.
★ For disinfecting drinking water.
★ For manufacturing chloroform $CHCl_3$

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Baking Soda ($NaHCO_3$) Sodium hydrogen carbonate

Preparation:
on reacting cold concentrated sodium chloride (brine) solution with ammonia and carbondioxide.

$$NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$$

brine — Ammonia — Ammonium chloride — Sodium hydrogen carbonate

Uses:-
For making baking powder, which is a mixture of baking soda and a mild edible acid such as tartaric acid. When baking powder is heated or mixed in water, this reaction takes place:-

$$NaHCO_3 + H^+ \longrightarrow CO_2 + H_2O + \text{Sodium salt of acid}$$

- Ingredient in antacids
- Uses in soda-acid fire extinguisher.

Washing Soda ($Na_2CO_3 \cdot 10H_2O$)

★ Recrystallisation of Sodium Carbonate gives Washing Soda.

Formation

$$NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$$

Ammonium chloride — Sodium hydrogen carbonate



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$2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$

Sodium hydrogen carbonate — Sodium carbonate

$Na_2CO_3 + 10H_2O \longrightarrow Na_2CO_3 \cdot 10H_2O$

Uses

- ★ Softening hard water
- ★ Washing clothes
- ★ In paper, paint and textile industry.
- ★ Manufacturing of sodium compounds like borax.

Q:- What is "$10H_2O$" in washing soda ?

Water of crystallisation, Actually the crystals of salt appears to be dry but actually they are not they contains water of crystallisation i.e the fixed number of water molecule

eg:- other examples ($CuSO_4 \cdot 5H_2O$ & $CaSO_4 \cdot 2H_2O$)

Plaster of Paris [Calcium sulphate hemihydrate, $CaSO_4 \cdot \frac{1}{2}H_2O$]

Preparation:

Heating gypsum at 373 K, Gypsum loses water molecules and becomes plaster of paris.

$$\underset{\text{Gypsum}}{CaSO_4 \cdot 2H_2O} \xrightarrow[\text{Heat}]{373K} \underset{\text{POP}}{CaSO_4 \cdot \tfrac{1}{2}H_2O} + 1\tfrac{1}{2}H_2O$$

Plaster of Paris is a white powder and on mixing with water, it changes to gypsum giving a hard solid mass.

$$\underset{\text{POP}}{CaSO_4 \cdot \tfrac{1}{2}H_2O} + 1\tfrac{1}{2}H_2O \longrightarrow \underset{\text{gypsum}}{CaSO_4 \cdot 2H_2O}$$

* Uses

- Used in hospital for setting fractured bone in right position.
- POP is used in making decorative items.
- Used in fireproofing material.
- For sealing Air-Tight gaps.